UltraXAI — Project Update Report

Explainable AI for Breast Cancer Detection from Ultrasound
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1. Summary

This update adds data augmentation to expand the training dataset, fixes a data-leakage bug that inflated earlier accuracy, adds modern training techniques to improve real accuracy, makes the code compatible with Albumentations 2.x, and provides a one-command training pipeline for a local GPU.

2. Data Augmentation & Dataset Volume

New script `training/data/augment_offline.py` writes N augmented copies of every training image and its mask. With 5 copies the training set grows 6×. Only the training split is augmented, so validation/test images never leak into training.

Augmentation	Purpose
Flips, shift / scale / rotate (Affine)	Probe position and orientation changes
Elastic & grid distortion	Tissue deformation, probe pressure
Speckle (multiplicative) & Gaussian noise, blur	Ultrasound noise variety
Brightness / contrast, gamma, CLAHE	Different machine gain settings
Sharpen, coarse dropout	Edge robustness, learn from multiple regions

2.1 Dataset Volume — Actual Numbers (BUSI)

BUSI contains 647 benign/malignant images (437 benign, 210 malignant). It was split 70/15/15, keeping the class ratio the same in every split. Only the training split was augmented, with 5 copies per image.

Split	Benign	Malignant	Total
Original dataset	437	210	647
Train (original)	305	147	452
Validation	66	31	97
Test (held-out)	66	32	98

Training images	Benign	Malignant	Total
Original train images	305	147	452
+ Augmented copies (5 per image)	1,525	735	2,260
Total images used for training	**1,830**	**882**	**2,712**

Result: the training set grew from 452 to 2,712 images, a **6× increase (+2,260 images, +500%)**. Validation (97) and test (98) images are kept original and unseen, so the reported scores are honest. Training also applies random augmentation each epoch, so the model sees new variations of all 2,712 images every epoch.

2.2 All Available Datasets

Dataset	Images
BUSI	780
BUS-BRA	1,875
UDIAT	163
STU	756
OASBUD	200
Total (raw)	**3,774**

Currently only BUSI is used (647 benign + malignant images; the 133 normal images are excluded), which gives 2,712 training images after augmentation. If all five datasets were used, about 2,640 images (70% of 3,774) would go to training, or **about 15,800 training images** after 6× augmentation (estimate).

3. Accuracy Improvements (train_classifiers.py)

Technique	Benefit
Stratified train / val split	Fixes bug: validation used the same images as training, so old scores were too high
Offline-augmented data (--aug_dirs)	More, more varied training data
Label smoothing 0.1	Better calibrated, less over-confident
Mixup (alpha 0.2)	Strong regularisation for small datasets
Warmup + cosine learning rate	Stable fine-tuning of pretrained models
AdamW weight decay 1e-4, gradient clipping	Less overfitting, stable training
Mixed precision (AMP)	Faster GPU training
Flip test-time augmentation	More robust predictions
Save-best fix	Model is saved even if first F1 is 0
Separate validation folder	Validation uses the separate 97-image val split, so no augmented copy of a validation image is ever trained on

4. Compatibility Fixes

Updated to the Albumentations 2.x API: ShiftScaleRotate → Affine, GaussNoise(var_limit) → std_range, new CoarseDropout ranges, removed OpticalDistortion shift_limit, and torch.amp.GradScaler.

5. One-Command Pipeline

`run_full_training.py`: (1) stratified 70/15/15 split → (2) augment train only → (3) train on 2,712 images and validate on 97 unseen images with ResNet50 + EfficientNetV2 → (4) save weights to backend/weights/ for the app.

```
python download_busi.py
python run_full_training.py --data_dir datasets/BUSI --copies 5 --epochs 80
```

6. Verification

The full pipeline was tested end-to-end (split, augment, 1 epoch per model, weights saved) on synthetic images with no errors or warnings. Final accuracy on the real BUSI dataset will be reported after GPU training. Note: scores are expected to be lower than before but honest, because the leakage bug is fixed.

7. Files Changed

File	Change
training/data/augment_offline.py	New — dataset expansion
training/data/transforms.py	New augmentations, Albumentations 2.x
training/train_classifiers.py	Split fix + accuracy techniques
run_full_training.py	New — one-command pipeline
README.md	Augmentation, accuracy, changelog, author credit

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